

Task 3: E-Commerce Web Application

Objective

Develop a **full-stack E-Commerce Web Application** that allows users to browse products, add items to a cart, place orders, and manage purchases. The application should include authentication, product management, and a database backend.

This project demonstrates practical skills in:

- Full Stack Web Development
 - Database Management
 - User Authentication
 - CRUD Operations
 - Session Management
 - Responsive UI Design
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Features Required

1. User Authentication

Users should be able to:

Register

Username

Email

Password

Confirm Password

Login

Email

Password

Logout

- Secure session handling
- Password hashing using bcrypt

2. Product Catalog

Display products with:

- Product Image
- Product Name
- Description
- Price
- Category
- Stock Availability

Example:

Wireless Mouse

Price: ₹599

Category: Electronics

Stock: Available

3. Shopping Cart

Users can:

- Add Products
- Remove Products
- Update Quantity
- View Cart Total

Example:

Product	Qty	Price
Mouse	2	₹1198

Keyboard	1	₹899
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Total: ₹2097

4. Order Management

Users can:

- Place Orders
- View Previous Orders
- Track Order Status

Statuses:

Pending
Processing
Shipped
Delivered

5. Admin Dashboard

Admin should be able to:

Product Management

- Add Product
- Edit Product
- Delete Product

Order Management

- View Orders
- Update Order Status

User Management

- View Registered Users
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Recommended Technology Stack

Frontend

- HTML5
- CSS3
- Bootstrap 5
- JavaScript

Backend

- Flask

Database

- SQLite

Authentication

- Flask-Bcrypt
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Database Design

Users Table

```
CREATE TABLE users(  
  id INTEGER PRIMARY KEY AUTOINCREMENT,  
  username TEXT,  
  email TEXT UNIQUE,  
  password TEXT  
);
```

Products Table

```
CREATE TABLE products(  
  id INTEGER PRIMARY KEY AUTOINCREMENT,  
  name TEXT,
```

```
description TEXT,  
price REAL,  
category TEXT,  
image TEXT,  
stock INTEGER  
);
```

Cart Table

```
CREATE TABLE cart(  
  id INTEGER PRIMARY KEY AUTOINCREMENT,  
  user_id INTEGER,  
  product_id INTEGER,  
  quantity INTEGER  
);
```

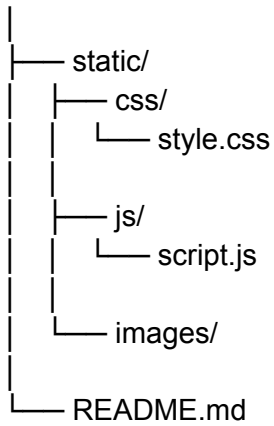
Orders Table

```
CREATE TABLE orders(  
  id INTEGER PRIMARY KEY AUTOINCREMENT,  
  user_id INTEGER,  
  total_amount REAL,  
  order_date DATE,  
  status TEXT  
);
```

Project Structure

ecommerce-web-app/

```
|  
|— app.py  
|— ecommerce.db  
|— requirements.txt  
|  
|— templates/  
|   |— login.html  
|   |— register.html  
|   |— home.html  
|   |— products.html  
|   |— cart.html  
|   |— checkout.html  
|   |— orders.html  
|   |— admin.html
```



Main Pages

Home Page

Display:

Welcome to ShopEase

Featured Products

Latest Deals

Categories

Product Page

Show all products.

Each product card:

Product Image

Product Name

Price

Add to Cart Button

Cart Page

Display:

Your Shopping Cart

Product Name

Quantity

Price

Remove Button

Total Amount

Checkout Page

Collect:

Name

Address

Phone Number

Payment Method

Orders Page

Display:

Order ID

Order Date

Total Amount

Order Status